CHEM 1301 Driller 9-3

Section 8.7 (empirical and molecular formulas) + Chapter 9 — 15 exam-style multiple-choice questions

Instructions. Choose the single best answer (a–e) for each question. You may use a periodic table. Calculators are allowed. Answers are rounded to the precision shown. Time yourself: aim for about 30 minutes (roughly 2 minutes per question, the same pace as the full exam).

Answer sheet. Use the bubble answer sheet (`CHEM1301/exam-prep/answer-sheet/answer-sheet.pdf`), rows 1–15.

1. When finding an empirical formula, after converting each element's mass to moles, you should next:

a) divide every element's moles by the molar mass of the compound
b) divide each element's mass by the smallest mass
c) divide every element's moles by the largest mole value
d) round each mole value to the nearest whole number
e) divide every element's moles by the smallest mole value

2. Sucrose has the molecular formula $C_{12}H_{22}O_{11}$. What is its empirical formula?

a) $C_6H_{11}O_{11}$
b) $C_{12}H_{22}O_{11}$
c) $C_{12}H_{22}O$
d) $C_6H_{11}O_5$
e) CH_2O

3. A 2.000 g sample of an iron oxide contains 1.447 g of iron; the rest is oxygen. What is its empirical formula?

a) Fe_3O_4
b) Fe_6O_8
c) FeO
d) Fe_3O
e) FeO_4

4. A white powder is 39.05% K, 1.01% H, 12.00% C, and 47.94% O by mass. What is its empirical formula?

a) K_2HCO_3
b) $K_{39}HC_{12}O_{47}$
c) $K_2H_2C_2O_6$
d) K_3HCO
e) $KHCO_3$

5. A manganese oxide is 49.55% Mn and 50.45% O by mass. What is its empirical formula?

a) Mn_2O_7
b) MnO
c) Mn_4O_{14}
d) MnO_3
e) MnO_7

6. Ribose has the empirical formula CH_2O and a molar mass of 150.13 g/mol. What is its molecular formula?

a) CH_2O
b) $C_5H_{10}O_5$
c) C_5H_2O
d) $C_4H_8O_4$
e) $C_6H_{12}O_6$

7. A liquid hydrocarbon is 92.26% C and 7.74% H by mass, and its molar mass is 78.11 g/mol. It burns completely in O_2 to form CO_2 and H_2O. How many moles of O_2 are needed to burn 0.400 mol of the hydrocarbon?

a) 0.400 mol
b) 0.500 mol
c) 2.40 mol
d) 3.00 mol
e) 6.00 mol

8. Which statement about the **excess** reactant in a reaction is correct?

a) It is used up before the limiting reactant
b) It determines the theoretical yield
c) Its leftover amount is added to the theoretical yield
d) It is always the reactant present in the larger mass
e) Some of it remains unreacted after the reaction stops

9. Octane burns: $2C_8H_{18}(l) + 25O_2(g) \rightarrow 16CO_2(g) + 18H_2O(g)$. How many moles of O_2 are needed to burn 0.300 mol of C_8H_{18}?

a) 0.0240 mol
b) 0.300 mol
c) 2.40 mol
d) 3.75 mol
e) 120 mol

10. Mercury(II) oxide decomposes when heated: $2HgO(s) \rightarrow 2Hg(l) + O_2(g)$. How many moles of O_2 form from 21.7 g of HgO?

a) 0.0501 mol
b) 0.100 mol
c) 0.200 mol
d) 1.60 mol
e) 4.99 mol

11. The thermite reaction is $Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s)$. What mass of Fe forms from 40.0 g of Fe_2O_3 with excess Al?

a) 0.501 g
b) 6.99 g
c) 14.0 g
d) 28.0 g
e) 80.0 g

12. Lithium burns according to the **unbalanced** equation $Li(s) + O_2(g) \rightarrow Li_2O(s)$. What mass of Li_2O forms from 3.47 g of Li with excess O_2?

a) 0.250 g
b) 1.74 g
c) 7.47 g
d) 14.9 g
e) 29.9 g

13. Silicon carbide forms by $SiO_2(s) + 3C(s) \rightarrow SiC(s) + 2CO(g)$. A mixture of 30.0 g of SiO_2 and 15.0 g of C is heated. Which statement is correct?

a) C is limiting, and 30.0 g of SiO_2 is left over
b) C is limiting, and 25.0 g of SiO_2 is left over
c) C is limiting, and 4.98 g of SiO_2 is left over
d) Neither is limiting; both reactants are used up completely
e) SiO_2 is limiting, and C is left over

14. For $3NO_2(g) + H_2O(l) \rightarrow 2HNO_3(aq) + NO(g)$, 23.0 g of NO_2 reacts with 5.00 g of H_2O. What is the theoretical yield of HNO_3?

a) 0.333 g
b) 21.0 g
c) 28.0 g
d) 31.5 g
e) 35.0 g

15. Yeast ferments glucose: $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$. Fermenting 45.0 g of glucose gives 19.5 g of ethanol, C_2H_5OH. What is the percent yield?

a) 0.847%
b) 43.3%
c) 84.7%
d) 118%
e) 169%